

Table 6. Descriptive statistics for price returns of observed commodities

Variables	Mean	SD	Min	Max	Skewness	Kurtosis	ADF test	JB test
Corn, China	-0.0026	0.3622	-2.3532	3.7373	1.0093	13.7628	0.0100	0.0000
Wheat, China	-0.0067	0.3739	-3.0912	2.9940	-0.6510	14.5726	0.0100	0.0000
Japonica Rice, China	-0.0087	0.3982	-2.7880	3.5148	1.0755	20.2388	0.0100	0.0000
Pork, China	-0.0094	1.0889	-5.9607	6.2092	0.0231	7.0155	0.0100	0.0000
Beef, China	0.0060	0.6964	-6.7499	6.6014	-0.1007	13.0687	0.0100	0.0000
Mutton, China	0.0095	0.7321	-2.6469	3.1276	-0.0118	3.4118	0.0100	0.0020
Cotton, China	-0.0024	0.5898	-4.6436	5.6136	0.6246	19.5942	0.0100	0.0000
Soybean, China	0.0118	0.4583	-4.7086	4.3450	0.9673	31.1252	0.0100	0.0000
Soybean Meal, China	-0.0192	0.9532	-4.5747	7.2667	1.0683	10.2258	0.0100	0.0000
Soybean Oil, China	-0.0185	1.1500	-5.9313	5.0498	-0.1693	5.8430	0.0100	0.0000
Rapeseed Meal, China	-0.0146	1.0220	-4.5847	6.4375	0.8465	10.2879	0.0100	0.0000
Rapeseed Oil, China	-0.0081	1.1400	-6.8115	5.3714	-0.0583	6.4320	0.0100	0.0000
Palm Oil, China	-0.0137	1.4858	-18.4807	7.4800	-1.2781	18.6321	0.0100	0.0000
Pure Benzene, China	-0.0026	1.4095	-10.7997	14.1219	0.2471	18.4464	0.0100	0.0000
Toluene, China	0.0039	1.1311	-9.2131	9.3000	-0.0070	13.4674	0.0100	0.0000
O-Xylene, China	0.0099	1.1023	-7.4203	11.1079	1.3232	23.7894	0.0100	0.0000
Styrene, China	-0.0175	1.6070	-10.3402	10.0560	0.2340	7.6834	0.0100	0.0000
Butadiene, China	-0.0123	2.2667	-12.3208	13.9373	0.0886	9.4869	0.0100	0.0000
P-Xylene, China	-0.0237	1.5376	-16.2442	8.5590	-0.9445	13.9687	0.0100	0.0000
PTA, China	-0.0145	1.3797	-8.2310	6.7398	-0.2393	7.3977	0.0100	0.0000
Epoxy Ethane, China	-0.0082	1.2057	-13.4930	7.7667	-3.1700	35.0657	0.0100	0.0000
Propylene, China	-0.0091	1.2744	-11.6015	15.2708	0.4893	23.6155	0.0100	0.0000
Polystyrene, China	-0.0230	1.0313	-5.2791	9.6255	1.5310	17.0096	0.0100	0.0000
Natural Rubber, China	-0.0232	1.5390	-8.4658	7.2001	0.0920	6.5314	0.0100	0.0000
PP, China	-0.0108	0.9297	-10.1726	5.6016	0.4474	17.8591	0.0100	0.0000
Causticsoda, China	0.0278	1.4504	-15.1509	16.6507	0.5168	38.9589	0.0100	0.0000
Potassic Fertilizers, China	0.0039	0.7050	-5.2137	7.5343	2.6148	29.2376	0.0100	0.0000
Xylene, China	-0.0036	1.1143	-10.3128	6.8215	-0.2493	15.8006	0.0100	0.0000
Glycol, China	-0.0194	1.4599	-9.8026	8.1706	-0.2094	7.8711	0.0100	0.0000
Acrylamide, China	0.0030	0.6792	-4.9206	4.7922	-0.3203	10.9859	0.0100	0.0000
Acrylic Acid, China	-0.0167	1.3941	-9.5470	10.6484	1.0855	16.2648	0.0100	0.0000
Acrylonitrile, China	-0.0059	0.9693	-6.0991	7.7739	0.3485	11.1583	0.0100	0.0000
Butyl Acrylate, China	-0.0131	1.2804	-8.2198	13.4476	1.7143	17.7077	0.0100	0.0000
PET Chips, China	0.0044	0.8512	-4.6584	6.4446	1.3152	15.1900	0.0100	0.0000
Methyl Methacrylate, China	0.0039	1.2062	-6.4812	15.0645	2.1437	26.6709	0.0100	0.0000
PVC, China	-0.0021	1.0006	-7.3834	9.3042	0.4139	14.1682	0.0100	0.0000
Urea, China	0.0142	0.8203	-10.6928	6.7441	-0.4612	26.2909	0.0100	0.0000
Methanol, China	0.0058	1.6079	-9.7630	14.9964	0.0952	10.9561	0.0100	0.0000
Metallurgical Coke, China	0.0354	1.0580	-15.0944	8.4233	-1.4787	32.9170	0.0100	0.0000
Hard Coking Coal, China	0.0325	0.7257	-3.8728	10.1456	2.3124	34.5219	0.0100	0.0000
Coking Coal, China	0.0372	0.6312	-4.1351	7.8409	2.3817	30.5122	0.0100	0.0000
Hot-rolled Sheet/Coil, China	0.0271	0.9769	-5.6261	13.4605	2.7393	38.4239	0.0100	0.0000
Platinum, China	-0.0262	1.5024	-10.5154	10.7739	-0.0351	7.6209	0.0100	0.0000
Silver, China	0.0092	1.6148	-11.7802	9.1510	-0.3517	10.0953	0.0100	0.0000
Palladium, China	0.0088	2.6602	-20.9868	23.4487	-0.0205	10.2605	0.0100	0.0000
Gold, China	0.0214	0.8488	-6.0288	4.9502	-0.2685	7.4191	0.0100	0.0000
Copper, China	0.0232	1.1078	-5.4348	7.9928	0.4035	6.8024	0.0100	0.0000
Nickel, China	-0.0382	1.8819	-22.5744	15.3835	-1.0013	19.6813	0.0100	0.0000
Aluminum, China	0.0231	1.1896	-6.8646	5.0430	-0.2821	4.7670	0.0100	0.0000
Zinc, China	0.0193	1.3507	-6.1240	7.8411	0.3482	5.0302	0.0100	0.0000
Lead, China	0.0218	1.0229	-4.8136	6.0535	0.1983	4.6305	0.0100	0.0000
Tin, China	0.0007	1.3001	-6.9560	6.9692	-0.2240	7.5769	0.0100	0.0000
Chrome Metal, China	-0.0052	0.8003	-7.1555	14.2742	4.6796	89.3530	0.0100	0.0000
Silicon Metal, China	0.0099	1.2452	-14.1143	24.3716	6.4960	154.9180	0.0100	0.0000
Cobalt, China	-0.0088	1.4091	-9.3622	12.8905	1.0417	17.5054	0.0100	0.0000
Ingot, China	0.0005	1.5967	-14.9195	34.2224	5.9898	143.8461	0.0100	0.0000
Electrolytic Manganese, China	0.0031	1.2694	-10.8963	12.8512	0.3357	32.3224	0.0100	0.0000
Nickel Sulfate, China	-0.0114	0.8934	-12.9355	12.4622	1.2762	71.2698	0.0100	0.0000
Antimony, China	0.0621	0.8233	-9.3976	10.7930	2.0336	43.8076	0.0100	0.0000
Soybean Oil, US	0.0318	2.1826	-32.8165	33.5687	-0.5165	72.6148	0.0100	0.0000
Diesel Oil, Singapore	-0.0138	2.4171	-22.5979	17.8139	-0.6204	13.7107	0.0100	0.0000
Gasoline, Japan	-0.0042	2.5816	-23.9706	28.5270	-0.4345	22.6123	0.0100	0.0000
Brent	0.0788	2.8917	-25.6389	41.2023	1.2893	43.4437	0.0100	0.0000
WTI	0.1130	3.1980	-28.1380	42.5830	1.6281	45.1246	0.0100	0.0000
Henry Hub Natural Gas	-0.0812	7.2570	-102.5103	143.2814	2.5289	116.5373	0.0100	0.0000
Crude Oil, UAE Dubai	0.0097	2.6704	-39.2108	29.7041	-1.3770	45.2310	0.0100	0.0000
Naphtha, Korea	0.0048	2.5645	-27.9156	28.4043	-0.4866	27.6755	0.0100	0.0000
Gold, UK	0.0306	0.8914	-5.1524	5.1334	-0.1965	6.2548	0.0100	0.0000
Silver, UK	0.0174	1.7326	-19.5856	10.2088	-0.6489	15.6982	0.0100	0.0000
Platinum, UK	-0.0123	1.5948	-14.4183	9.3411	-0.4359	8.7199	0.0100	0.0000
Palladium, UK	0.0448	2.3871	-15.6769	16.9615	-0.5041	10.0149	0.0100	0.0000

Note: SD represents the standard deviation. ADF is for the p -value of augmented Dickey-Fuller test for unit root, with alternative hypothesis of being a stationary series. JB is for the p -value of empirical statistic of the Jarque-Bera test for normality, with null hypothesis of being normal.